Student Print Packet

EBIKE101 | CTS Cohort 6 | Day 33 | Monday, August 31, 2026

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EBIKE101 Day 1 Student Packet

Safe micro-mobility starts before the throttle moves.

Name: _____ Team: _____

Today's big question

What makes an e-bike different from a bicycle, and what must be proven safe before anybody rides it?

Part 1: Bicycle baseline

Label or describe each part. Use plain language.

Part	What does it do?	What could make it unsafe?
Frame		
Fork		
Front wheel		
Rear wheel		
Tires		
Brakes		
Chain/drivetrain		
Handlebars/stem		
Seat/seatpost		
Pedals/cranks		

Part 2: E-bike added systems

E-bike part	Job	What could fail?
Battery		
Battery mount		
Motor		
Controller		
Display		
Throttle		
Pedal assist sensor		
Wiring harness		
Connectors		
Charger		
Brake cutoffs, if installed		

Part 3: Basic system flow

Fill in the flow.

Rider input -> _____ -> battery power -> _____ -> wheel motion

What does the rider control?

_____

What does the controller control?

_____

What does the motor move?

_____

Part 4: Pre-ride inspection checklist

Mark each item.

P = Pass

F = Fail

R = Needs instructor review

N/A = Not installed or not available today

Mechanical bike

Item	P/F/R/N/A	Notes
Frame has no obvious cracks, damage, or loose parts		
Fork looks straight and secure		
Wheels are secure		
Tires have usable pressure and no obvious damage		
Brakes squeeze firmly and stop the bike during a roll check		
Handlebars and stem are tight		
Seatpost is secure		
Chain/drivetrain is present, routed, and not jammed		
Pedals/cranks are secure		

Electrical assist system

Item	P/F/R/N/A	Notes
Battery case has no swelling, heat, odor, or visible damage		
Battery is mounted securely		
Controller is mounted securely		
Motor cable is protected and not rubbing		
Connectors are seated and not damaged		
Wiring is clear of tires, chain, brakes, cranks, and pinch points		

Display/control unit is secure		
Throttle/PAS is not being touched during setup		
Charger is known, matched, and undamaged		

Rider and area

Item	P/F/R/N/A	Notes
Helmet available and fits		
Controlled area can be marked before any ride test		
No public traffic or uncontrolled pedestrian path		
Instructor has approved the next test step		

Part 5: Team risk report

Write one risk in each category.

Mechanical risk:

_____

Electrical risk:

_____

Rider-control risk:

_____

Area/environment risk:

_____

Exit ticket

1. What makes an e-bike different from a normal bicycle?

_____

2. What are three things that must be checked before any powered test?

_____

3. What is one thing you will check first tomorrow?

_____